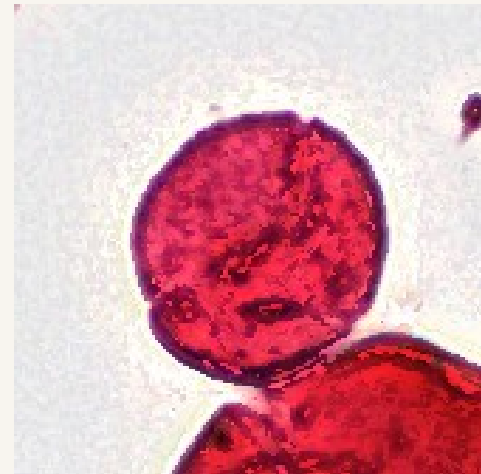


Identifying Fossil Pollen with Machine Learning

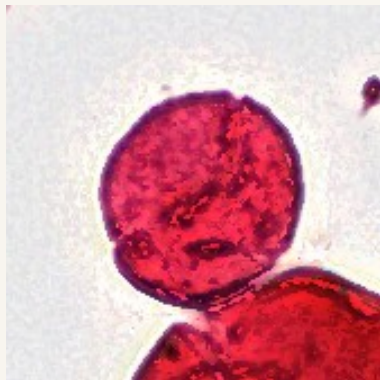
Help researchers identify fossil pollen categories more efficiently.



Patrick Batsell · Yun Ying Tsai · Yeonju Kim · Yunfan Bao · Alan Yang

What grew in North America during a warmer past?

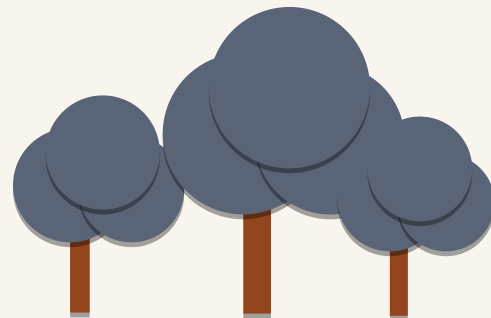
Fossil pollen preserves clues from about 50 million years ago.



Fossil pollen



Identify plant groups

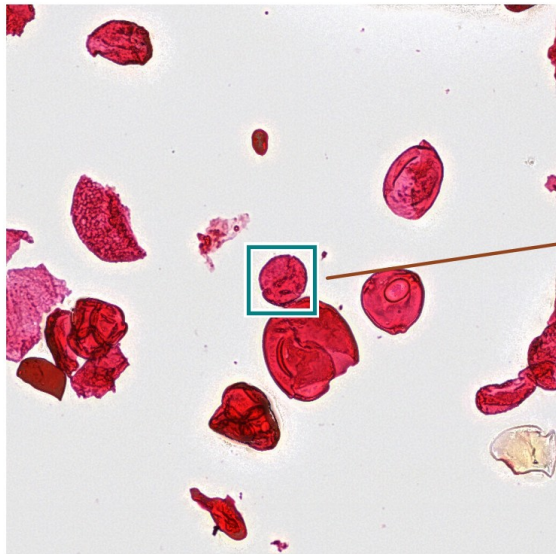


Reconstruct vegetation

The bottleneck: identifying thousands of specimens by hand.

Finding pollen is only the first step

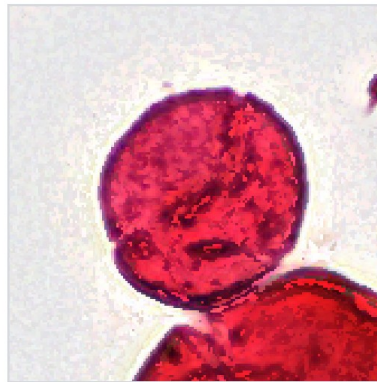
A A crowded microscope image



DETECTION

Where is the specimen?

B Inspect the specimen



Example expert label

Bombacacidites sp.

CLASSIFICATION

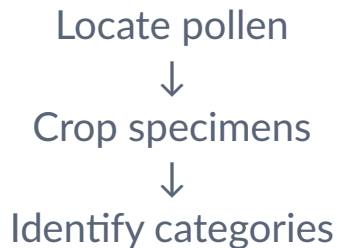
What type is it?

Our goal: assign a category to each detected specimen.

What earlier studies suggest

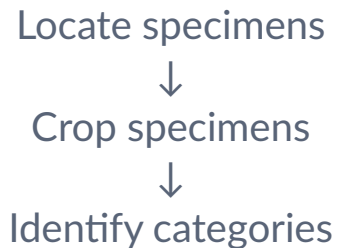
01

Punyasena et al.



02

Martinsen et al.



03

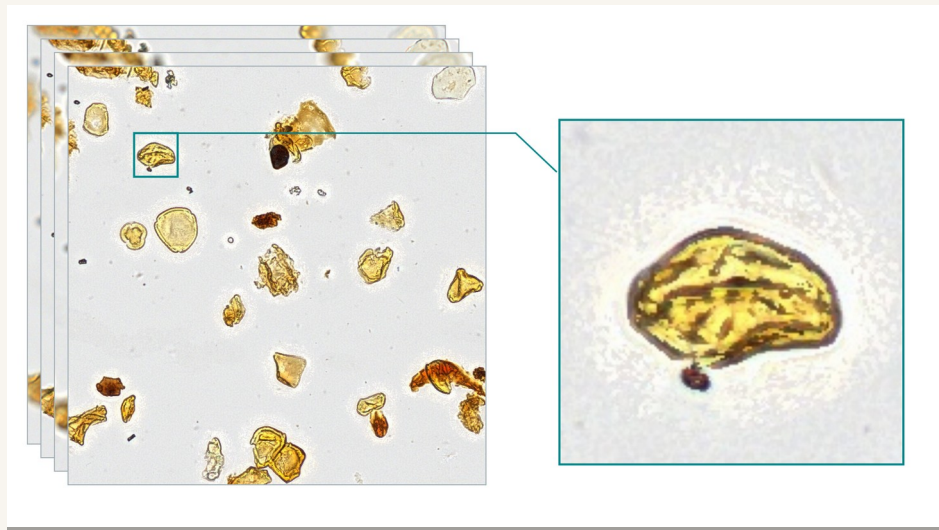
Abbas Shaikh et al.



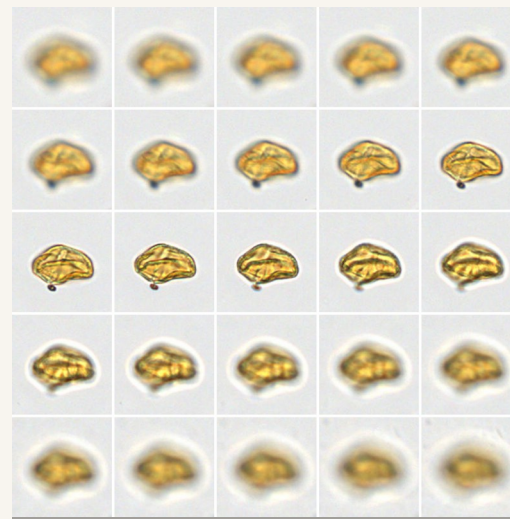
Build on whole-slide detection; add category identification.

One specimen, multiple focal views

- An expert-annotated pollen grain



This example: 25 focal planes
About 30 GB for the full scan



Some categories have far fewer examples

83

image-annotation pairs

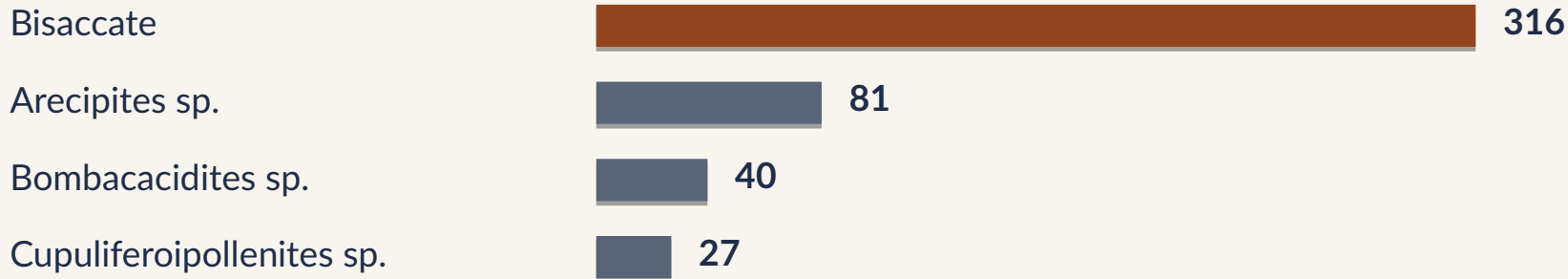
32

priority categories

2,290

priority annotations

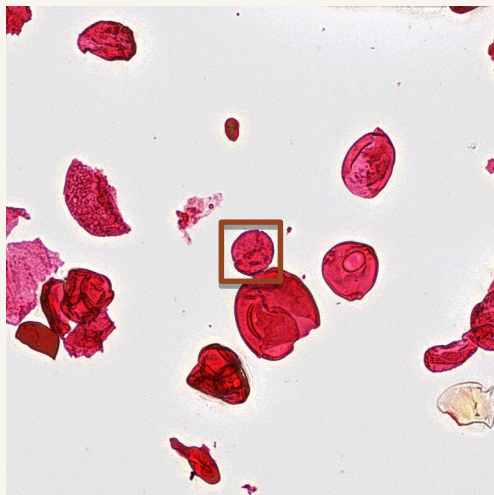
LABELED EXAMPLES • FOUR SELECTED CATEGORIES



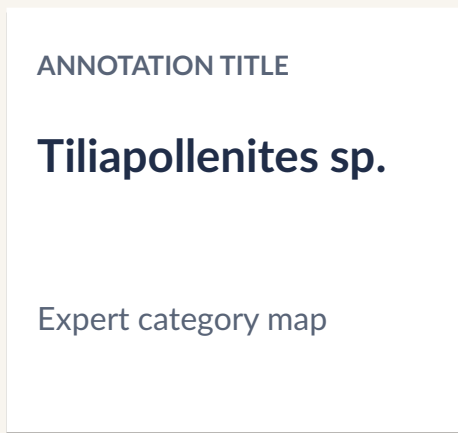
Fewer examples can make a category harder to learn.

Turn expert annotations into learning examples

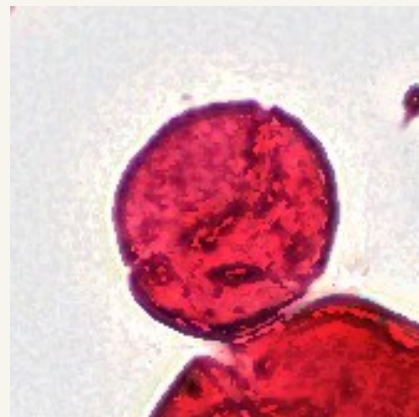
1 Locate the specimen



2 Read its expert label



3 Extract a labeled crop



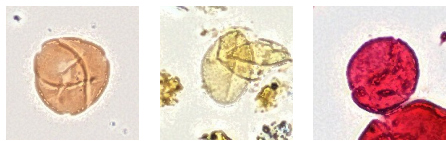
Bombacacidites sp.

Check that the crop is clear and contains the marked specimen.

Learn from some slides; test on unseen slides

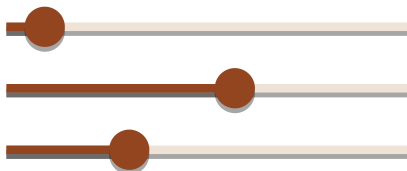
Mix up the microscope slides, then separate them into three groups.

Training



Learn from examples

Validation



Make adjustments

Testing



Check new slides

All examples from one microscope slide stay in the same group.

Does a dedicated classifier improve identification?

A ONE MODEL DOES BOTH JOBS

Find + identify

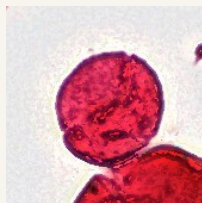
RF-DETR

→ Specimen locations + categories

B TWO MODELS SHARE THE WORK

Find specimens

Detector • RF-DETR



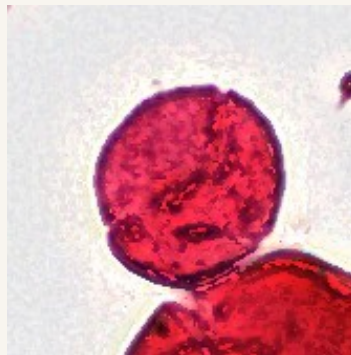
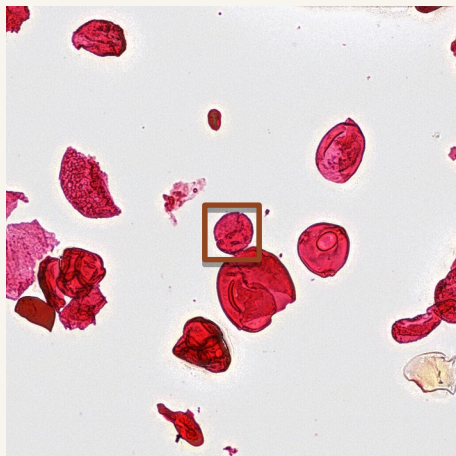
Identify each crop

Classifier • Swin-Tiny

Compare correct locations and labels on the same held-out slides.

Make fossil pollen collections easier to study

INTENDED OUTPUT: LOCATED, LABELED SPECIMENS FOR EXPERT REVIEW



A suggested category
for the expert to review

Example: *Bombacacidites* sp.

Illustration uses an expert label, not a model prediction.

Less manual identification. More evidence about past vegetation.

NEXT: prepare crops → train both approaches → compare their errors